		Target engagement	Dose	Efficacy	Safety
#6	**Alzheimer disease** HRH3 antagonists (3)	**2014** Pig PET and in-vitro H3 affinity of GSK239512 (vs human [11C]GSK189254 PET)	**2014** Rat five-trial inhibitory… Target exposure reached; ADAS-Cog failed	**2016** Rodent procognitive battery… Three rodent successes, three AD failures	**2016** Rat safety pharmacology an… Sponsor confirms dose-dependent sleep disturbance
	Alzheimer disease HTR6 antagonists (3)	**2023** Non-human primate PET 5-HT6 receptor occupancy… Rat under-predicted human brai…	**2022** Rat cognition battery (object… Rat efficacious exposure reached; trial failed	**2022** Rat cognition battery (object… Rat efficacy did not translate; Phase 2 null	**2014** IND-enabling toxicology (species and findings unpublished; no NDA, so no FDA review)
	Alzheimer disease anti-APP antibodies (3)	**2021** tg-ArcSwe mice, mAb158 Mice: no plaque fall; patients: PET fell	**2014** hAPP/PS1 mice, crenezumab 15 mg/kg failed; higher doses failed too	**2014** hAPP/PS1 mice, crenezumab Crenezumab missed both primaries	**2017** Aged APP mice given 2H6 (ponezumab parent) Excess microbleeds missed
#7	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2020** Eosinophil DP2 potency Target reached; exacerbations unchanged	**2014** Allergen challenge Challenge said yes; FEV1 trials missed	**2012** Off-target selectivity panels; animal toxicology unpublished
	asthma ADRB2 agonists (4)	**2017** Tanabe-era in vitro receptor-binding kinetics of TA-2005 (carmoterol):…	**2014** AstraZeneca guinea-pig histamine bronchoconstriction,… 2 ug once daily missed 24-h…	**2014** AstraZeneca guinea-pig histamine bronchoconstriction,… 24-h trough FEV1 primary missed	**2017** Guinea-pig isolated right atrium, left atrium and papillary muscle chronotropic/inotropic assa…
	asthma NR3C1 agonists (4)	**2017** Rat allergic airway inflammation (lung oedema; thymic involution as systemic index)	**2020** Rat allergic airway inflammation (lung oedema; thymic… Top dose missed week-12 FEV1	**2018** Allergen-induced late asthmatic response in mild atopic… Half-ICS LAR effect; 1 mg trial failed	**2020** Rat allergic airway inflammation (lung oedema; thymic involution as systemic index)
	asthma anti-IL13 antibodies (4)	**2013** IL-13-stimulated bronchial epithelial cells and patient brushin… Airway eosinophils did not fall	**2018** Integrated population PK/PD dose-exposure-FEV1 model of… Model dose reached; phase 3 failed	**2018** IL-13-humanised… Phase 3 exacerbation and OCS primaries missed	**2022** Cynomolgus repeat-dose toxicology (tralokinumab, lebrikizumab)
#8	**breast cancer** CDK4/6 inhibitors (6)	**2016** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Abemaciclib xenograft PK/PD (pRb to growth) Brain exposure OK, no response	**2014** Human breast cancer cell-line panel (47 lines) Brain-metastasis trial missed	**2017** Dog cardiovascular telemetry (QTc) with hERG assays Missed liver injury and clots
	breast cancer pan-HER inhibitors (3)	**2014** ER+/HER2-negative tamoxifen-resistant MCF7 and T47D lines and MCF7/TamRes xenograft…	**2017** HER2-amplified breast lines and xenografts (BT-474, SK-BR-3)	**2019** HER2-amplified breast lines and xenografts (BT-474, SK-BR-3)	**2017** HER2-amplified breast lines… Diarrhoea DLT despite stated animal window
#9	**osteoarthritis** MAPK14 inhibitors (3)	**2012** Rat iodoacetate joint degeneration + carrageenan thermal hyperalgesia (SB-203580, VX-745)	**2009** Rat and cynomolgus endotoxin-induced TNF-alpha PD model (PH-797804)	**2012** Rodent rheumatoid-arthritis models with ex-vivo LPS whole-blood cytokine PD…	**2014** Rat repeat-dose intra-articular toxicology (FX005)
	osteoarthritis NTRK1 inhibitors (3)	**2019** Rat MIA and MNX knee OA (AR786, sibling TrkA inhibitor)	**2019** Rat MIA and MNX knee OA (AR786, sibling TrkA inhibitor)	**2019** Rat PGPS knee arthritis flare (GZ389988, single intra-articular… Knockout says TrkA block enoug…	**2021** Rat 4- and 13-week oral neurotoxicity (ASP7962) Rat silent; two-thirds got joint…
#10	**colorectal cancer** KDR (VEGFR) inhibitors (3)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2017** Human colorectal patient-derived xenografts (PDX) in mice	**2023** Dog cardiovascular safety… Dogs silent; 14% grade 3-4 hypertension
#11	**chronic hepatitis B virus infection** HBV RNA antisense / siRNA (3)	**2022** Chronically HBV-infected chimpanzees	**2018** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)	**2023** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2023** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)
#12	**hepatocellular carcinoma** VEGFR multikinase inhibitors (4)	**2015** Human HCC cell-line and patient-derived xenografts in immunodeficient mice	**2018** Population PK / exposure-response model of lenvatinib in HCC (phase 1 pool +…	**2020** Mouse HCC with diverse genetic drivers (hydrodynamic transposon delivery) screened for…	**2015** GLP repeat-dose toxicology in rat plus dog or monkey
#13	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2019** Rodent occupancy dosing Predicted-adequate exposure, no effect	**2017** Ketamine challenge Challenge passed; phase 2 PANSS missed	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
#14	**gastric cancer** PARP1/2 inhibitors (3)	**2023** Gastroesophageal adenocarcinoma PDX platform and PDX-derived primary cells, BRCA2…	**2023** Gastroesophageal adenocarcinoma PDX platform and PDX-derived primary cells, BRCA2…	**2026** Gastroesophageal adenocarcinoma PDX platform an… Hypoxia-sensitisation bar misse…	**2017** Rat and dog general toxicology, up to 26 weeks (olaparib)
#15	**hepatitis C** HCV NS3/4A protease inhibitors (13)	**2013** Chronically HCV-infected chimpanzee (genotype 1a/1b)	**2009** Replicon EC99 vs rat plasma and liver exposure	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology
#17	**acute myeloid leukaemia** FLT3 inhibitors (4)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2012** Ba/F3 FLT3-ITD resistance screen Combination: no survival gain	**2018** Rat and dog toxicology and safety pharmacology PRES and QT missed by tox species
#19	**MASH** NR1H4 agonists (4)	**2019** Rodent FXR target-gene PD (tropifexor)	**2019** STAM and AMLN-diet NASH mice (tropifexor)	**2021** STAM and AMLN-diet NASH mice (tropifexor) Outcome trial negative	**2024** High-fat-diet mouse ALT/cholesterol screen Cilofexor: no lipid change
	MASH FGF21 analogue (FGFR1c/KLB)…	**2023** Obese/diabetic cynomolgus monkeys	**2023** Pegozafermin population PK/PD exposure-response model… Reached, failed fibrosis endpoin…	**2023** Pegozafermin population PK/PD exposure-response model… Pegbelfermin failed fibrosis…	**2020** Diet-induced obese mice given recombinant human FGF21… Mouse bone loss not seen at 16…
#20	**prostate cancer** AR antagonists (5)	**2013** LNCaP/AR castrate xenografts	**2013** LNCaP/AR castrate xenografts	**2016** LNCaP/AR castrate xenografts	**2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide)
#22	**ovarian cancer** PARP1/2 inhibitors (6)	**2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib)	**2021** NOVA exposure–safety model (niraparib)	**2016** BRCA-deficient cancer cells and xenografts (niraparib)	**2017** Dog cardiovascular safety pharmacology (niraparib)
#23	**rheumatoid arthritis** BTK inhibitors (8)	**2013** BTK occupancy probe in CIA mice	**2022** CIA occupancy-efficacy PK/PD Target reached, no effect	**2019** Collagen-induced arthritis (rodent) CIA-active drug failed Phase 2	**2021** Human platelet aggregometry and humanized-mouse thrombus formation (acalabrutinib vs ibrutinib)
	rheumatoid arthritis SYK inhibitors (3)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) Target reached, DAS28 unmove…	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) DAS28 no better than placebo	**2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib) Serious infections missed,…
#24	**multiple myeloma** TNFRSF17-directed depleters (8)	**2016** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2014** BCMA+ myeloma xenografts in mice, ADC	**2019** bb2121 CAR T cytotoxicity against cell lines of quantified BCMA density (in vitro)	**2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin)
#27	**open-angle glaucoma** ROCK inhibitors (3)	**2018** Living mouse eye, OCT of conventional outflow tissue, topical netarsudil	**2015** Normotensive monkey eye, topical netarsudil 0.04%	**2015** Normotensive monkey eye, topical netarsudil 0.04% Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
All.	**allergic rhinitis** allergen (immunotherapy) (8)	**2018** Conjunctival provocation test (CPT), gp-ASIT+	**2019** Conjunctival provocation te… CPT dose reached, confirmatory trial missed	**2019** Conjunctival provocation te… Confirmatory field trial missed primary	**2019** Allergen exposure chamber (EEC)
Autoimmune	**psoriasis** RORC inverse agonists (4)	**2021** Human whole-blood IL-17 secretion assay No IL-17 change in plaques	**2021** Human whole-blood IL-17 secretion assay	**2022** IL-23 minicircle mouse, oral BI 730357	**2015** Nonclinical toxicology species (not public)
	psoriasis anti-IL17A antibodies (4)	**2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** Cynomolgus repeat-dose toxicology	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** People with inborn errors of IL-17 immunity
	ulcerative colitis S1PR1 agonists (3)	**2016** Rodent blood lymphocyte counts (ozanimod)	**2023** Rat and dog repeat-dose toxicology (etrasimod)	**2016** TNBS and CD4+CD45RBhi transfer colitis mice (ozanimod)	**2023** hERG current assay in vitro (etrasimod) Ozanimod still slows the heart
GI	**GERD** ATP4A blockers (3)	**2019** Stimulated gastric acid secretion in dogs	**2017** Stimulated gastric acid secretion in dogs	**2019** Rat reflux oesophagitis vs esomeprazole	**2025** Rat repeat dosing: gastrin, emptying, microbiota
Hematology	**hemophilia A** F8 replacement / gene therapy (8)	**2025** Cynomolgus macaques with pre-existing anti-AAV5 antibodies given valoctocogene roxaparvovec…	**2017** F8-knockout mice + cynomolgus macaques	**2021** Hemophilia A dogs (Chapel Hill / Queen's colonies) Mouse-stable BAX 888 lost in all…	**2023** Hemophilia A dogs (Chapel Hill / Queen's colonies)
	hemophilia B F9 replacement / gene therapy (7)	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua FIX expression lost in 7 of 8	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua	**2024** Juvenile hemophilia B dogs, canine surrogate vector	**2024** Cynomolgus GLP tox, fidanacogene predecessor